In order not to violate Condition 2, Row 2 Column 1 = Row 3 Column 1 = 0.

$$\begin{bmatrix} 0 & \square \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \text{The corresponding matrix}$$

Find the general solutions of the systems whose augmented matrices are given in Exercises 7–14

7. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$

$\boxed{\text{Step 1:}}$ Row 2 = Row 2 − 3 × Row 1

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{bmatrix}$$

$\boxed{\text{Step 2:}}$ Row 2 = Row 2 × $\left(-\frac{1}{5}\right)$

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

The corresponding equations are:

$x_1 + 3x_2 + 4x_3 = 7$ (1)

$x_3 = 3$ (2)

$\therefore x_1 + 3x_2 + 4 \times 3 = 7$

$\Rightarrow \boxed{x_1 = -3x_2 - 5}$

The general solution is

$$\begin{cases} x_1 = -3x_2 - 5 \\ x_2 \text{ is free} \\ x_3 = 3 \end{cases}$$

9. $\begin{bmatrix} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -6 \end{bmatrix}$

$\boxed{\text{Step 1:}}$ Interchange Row 1 and Row 2

$$\begin{bmatrix} 1 & -2 & 7 & -6 \\ 0 & 1 & -6 & 5 \end{bmatrix}$$

$\boxed{\text{Step 2:}}$ Row 1 = Row 1 + 2 × Row 2

$$\begin{bmatrix} 1 & 0 & -5 & 4 \\ 0 & 1 & -6 & 5 \end{bmatrix}$$

The corresponding equations are:

$x_1 - 5x_3 = 4$ (1)

$x_2 - 6x_3 = 5$ (2)

$\therefore x_2 = 6x_3 + 5$ and $x_1 = 4 + 5x_3$

The general solution is:

$$\begin{cases} x_1 = 4 + 5x_3 \\ x_2 = 5 + 6x_3 \\ x_3 \text{ is free} \end{cases}$$